U.S. Software

The Rule of 40 Handbook - A Better Way To Judge Quality in Software

CORE

One approach to do well in these volatile times could be to focus on high quality stocks. The Rule of 40 is one method to identify such names, for which we do a deep dive within. SNOW, DDOG and ZI show the highest results. However, if valuation is also considered, CRM, ESTC and JAMF look more interesting.

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I think this is an important piece of info

Investment Summary:

Software, and growth assets more broadly, have seen a significant correction in the last 6 months and investors are increasingly looking to higher quality assets to weather the volatility. The Rule of 40 is thought of as an acclaimed KPI within software to identify such assets since it considers the trade off between growth and profitability. In this report, we are offering a deep dive into this KPI and its nuances. We can see that firms define the rule differently. Typically, software vendors use revenue growth, with some choosing to represent top-line growth with subscription revenue, ARR or ACV, while adj. EBITDA, operating profit and free cash flow (FCF) tend to be the most common proxies for profitability used in the calculation.

These nuanced definitions are not inherently wrong, but can result in materially different Rule of 40 calculations - up to 25% in certain cases. To be able to compare names in the sector, investors need to harmonize the calculation. We show within the report that FCF seems to show better results than EBITDA and operating profit. Using this approach we can see that SNOW, DDOG and ZI are the stocks with the highest Rule of 40 numbers.

To understand drivers for different outcomes we present five factors (deployment model, capital intensity, M&A, leverage and amortization of deferred costs) that must be considered when using the rule to assess operational quality.

Since a high quality company does not necessarily need to be a good investment, we next compare different variations of the Rule of 40 to valuations across our coverage. Our analysis of 15 different variations of the Rule of 40 shows that a modified approach, giving a higher weight to sales growth in the calculation, provides more explanatory power of valuation than the traditional metric. We conclude from our analysis that investors continue to appreciate growth, but margin, particularly FCF margin, should also be considered when seeking to identify undervalued assets. Using this framework, we identify ESTC, CRM and JAMF as stocks that look more interesting.

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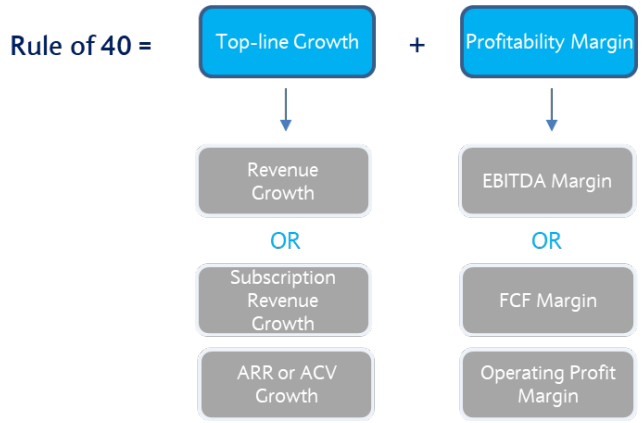
Rule of 40 Background

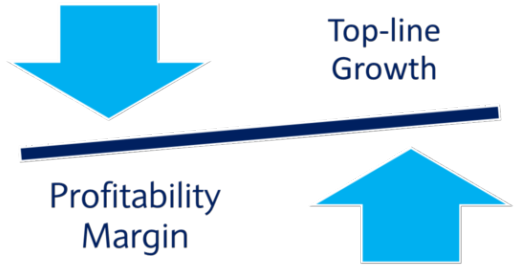
The 'Rule of 40' or 'Rule of 40%' is a metric that was originally conceived to provide venture capitalists with a rule of thumb to assess a healthy software company, and is touted as particularly useful since it provides a framework to compare companies at different stages of growth and maturity. The metric is calculated by adding together a top-line growth metric (most commonly revenue growth, but sometimes ARR or ACV growth is used as well) with a profitability metric (typically either adjusted EBITDA, operating margin, or FCF margin). The idea behind the metric (and name) is that companies that achieve a growth rate plus profitability margin above 40% represent healthy software businesses and companies which have a higher Rule of 40 should be more valuable. This is an especially powerful metric as it blends growth and profitability, which are the two greatest focuses for software companies and why most of our coverage is valued on either sales or cash flow.

FIGURE 1. **Rule of 40 Definition**



Source: Barclays Research

FIGURE 2. **Rule of 40 Considers Growth and Margin**



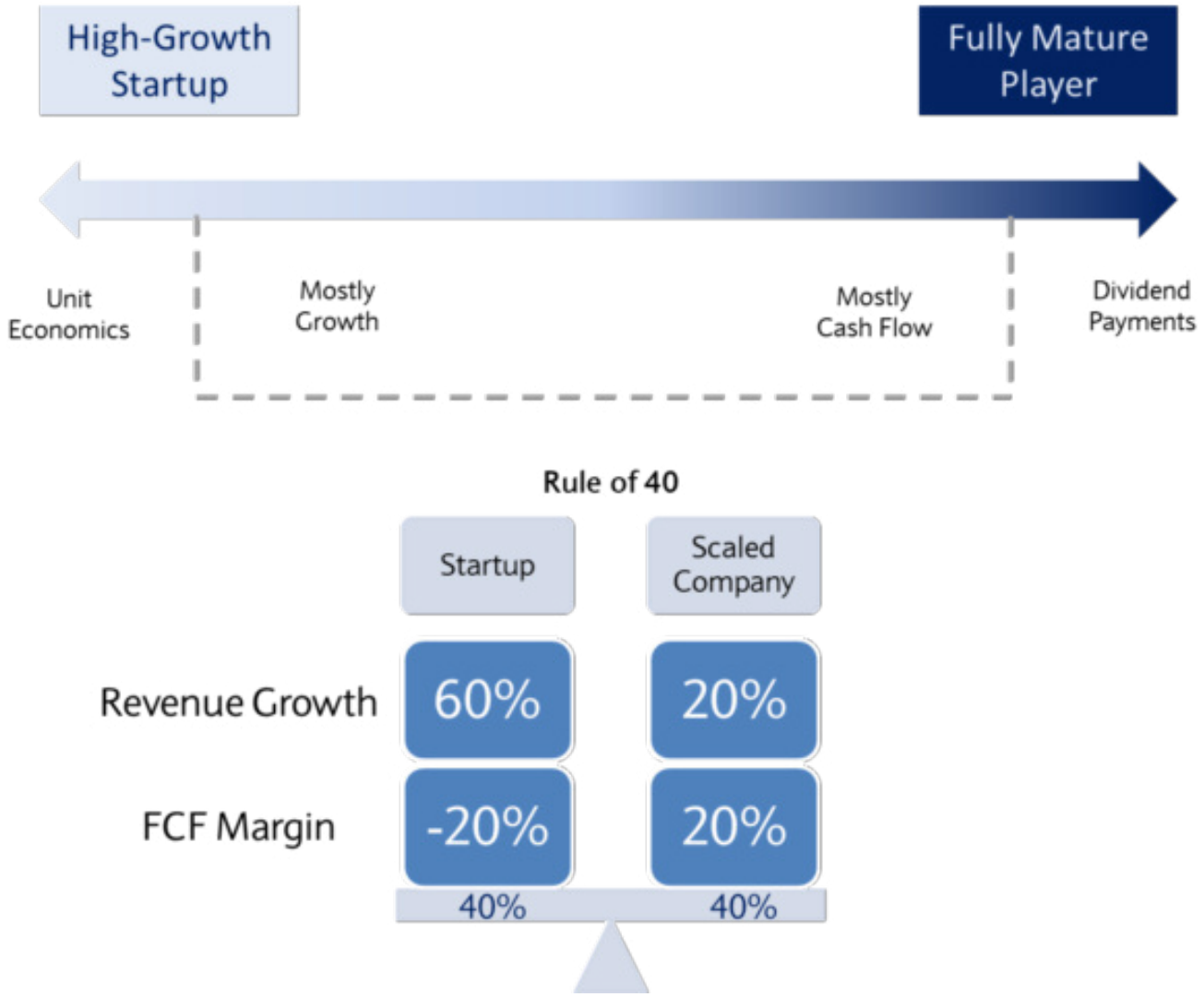
Source: Barclays Research

Benefits of Rule of 40

By most accounts, the Rule of 40 was popularized in two separate 2015 blog posts by Venture Capitalists Brad Feld and Fred Wilson, who attended the same board meeting where a late stage investor described what his firm called the "40% rule for a healthy software company", including SaaS companies. The greatest advantages of the Rule of 40 metric is the broad applicability and flexibility of use, given the ability to sub out the metrics used in summation coupled with the ability for a company to achieve the Rule of 40 in many different ways.

Near Universal Gauge for Holistic Performance: Due to the metric combining both top-line growth and profitability, there is a broad appeal for its use. Essentially, a SaaS start-up with 60% growth and -20% cash burn is treated as comparable to a more mature software company with 20% growth and a 20% FCF margin (see below). This makes sense, as a young company needs to invest to fuel growth and a mature company would likely trade off some growth for stronger profitability and has greater expectation of providing capital returns to investors in the form of share repurchases or dividends. The metric appreciates and normalizes for regular difference in scale, making it nearly universal.

FIGURE 3. **Rule of 40 Provides a Structure for Comparing Companies of Different Maturities**



Source: Barclays Research

Flexibility on Exact Chosen Metrics: Something else that adds to the broad use of the Rule of 40 is the ability to choose the metrics in the computation that make the most sense. This is allowed because there is no cut and dry formula for it, and it is more reliant on the concept of growth and profitability combining together, than one specific formula. For example, ARR growth can be used to normalize for a company going through a subscription transition, or unlevered FCF could be used to adjust for one company with a large debt component. Later in this report, we identify five factors that result in material differences when calculating the Rule of 40 that are important to understand when using the metric.

Rule of 40 Can be Materially Different Depending on Definition

As alluded to, the Rule of 40 is thought of as an acclaimed KPI since it considers the trade off between growth and profitability, however, firms define the rule differently, and the metric can produce, in some cases, materially different outcomes depending on the definition. To help make this more tangible, we present below examples of how public software vendors define or have referenced the Rule of 40.

FIGURE 4. **Rule of 40 Definitions Are Not Standardized**

Company	Rule of 40 Definition
ADBE	Rev. growth + non-GAAP operating margin
COUP	Rev. growth + adj. FCF margin
CRWD	Rev. growth + EBIT/FCF Margin
CYBR	Rev. growth + FCF margin
DH	Rev. growth + uFCF margin
DOCN	Rev. growth + adj. FCF margin
DT	ARR growth + adj. FCF margin
FTNT	Rev. growth + non-GAAP Operating Margin
MNDT	Rev. growth + non-GAAP Operating Margin
MSP	Rev. growth + adj. EBITDA margin
NOW	Sub rev. growth + adj. FCF margin
PATH	ARR growth + adj. FCF margin
PAYC	Rev. growth + adj. EBITDA margin
PEGA	ACV growth + adj. FCF margin
PTC	ARR growth + non-GAAP Operating margin
SPT	Rev. growth + adj. FCF margin
TENB	Rev. growth + adj. uFCF margin
ZS	Rev. growth + non-GAAP Operating Margin

Source: Barclays Research, Company data

While there is variety among proxies used for growth and profitability, we do not inherently think that this is wrong, but note in many cases, using different profitability metrics can lead to materially different conclusions. The most extreme examples tend to be the difference between Rule of 40 when using EBITDA margin compared to FCF margin, which in some cases, such as with DigitalOcean and Oracle, can lead to +25% differences in calculated “Rule of 40.” Below, we calculate the three most common Rule of 40 definitions across our coverage using total revenue growth summed with Adj. EBITDA margin, Adj. Operating Profit margin and FCF margin.

FIGURE 5. Rule of 40 Varies Greatly by Definition

Company Info		Rule of 40			Comparison	
Company	Ticker	Adj. EBITDA	Adj. Operating Profit	Free Cash Flow	Best Metric	Difference from Best to Worst
DigitalOcean	DOCN	67.0%	46.4%	40.2%	Adj. EBITDA	26.8%
Oracle	ORCL	56.2%	51.9%	30.2%	Adj. EBITDA	26.1%
Paycom	PAYC	65.2%	58.7%	44.3%	Adj. EBITDA	21.0%
Coupa	COUP	66.6%	46.2%	55.2%	Adj. EBITDA	20.3%
Ceridian	CDAY	37.4%	30.0%	20.1%	Adj. EBITDA	17.3%
OpenText	OTEX	41.3%	38.8%	24.1%	Adj. EBITDA	17.2%
Microsoft	MSFT	70.2%	63.2%	53.5%	Adj. EBITDA	16.7%
Qualtrics	XM	48.1%	43.8%	31.5%	Adj. EBITDA	16.6%
SAP	SAP	35.5%	31.4%	20.3%	Adj. EBITDA	15.2%
DoubleVerify	DV	69.2%	69.2%	58.4%	Adj. EBITDA	10.9%
UiPath	PATH	56.4%	55.4%	39.6%	Adj. EBITDA	16.8%
Lightspeed POS	LSPD	58.8%	56.6%	44.5%	Adj. EBITDA	14.3%
Intuit	INTU	74.1%	69.2%	57.9%	Adj. EBITDA	16.2%
Jamf	JAMF	54.3%	41.6%	51.7%	Adj. EBITDA	12.7%
Splunk	SPLK	15.4%	11.7%	24.0%	Free Cash Flow	12.3%
BigCommerce	BIGC	26.7%	23.9%	14.5%	Adj. EBITDA	12.2%
Confluent	CFLT	23.5%	22.5%	34.6%	Free Cash Flow	12.0%
Sprout Social	SPT	46.0%	37.9%	48.7%	Free Cash Flow	10.8%
Snowflake	SNOW	105.2%	103.4%	112.8%	Free Cash Flow	9.4%
Datadog	DDOG	88.3%	86.5%	94.8%	Free Cash Flow	8.3%
Integral Ad Science	IAS	66.4%	66.3%	58.3%	Adj. EBITDA	8.0%
ServiceNow	NOW	63.6%	55.6%	61.0%	Adj. EBITDA	8.0%
Teradata	TDC	32.5%	24.9%	26.9%	Adj. EBITDA	7.6%
WalkMe	WKME	6.9%	4.4%	11.3%	Free Cash Flow	6.9%
ZoomInfo	ZI	100.5%	97.8%	93.7%	Adj. EBITDA	6.8%
Workday	WDAY	48.1%	41.4%	46.0%	Adj. EBITDA	6.7%
Freshworks	FRSH	47.3%	43.7%	50.2%	Free Cash Flow	6.5%
Salesforce	CRM	49.7%	43.3%	44.6%	Adj. EBITDA	6.3%
VMware	VMW	46.0%	39.7%	40.1%	Adj. EBITDA	6.3%
Pegasystems	PEGA	27.2%	21.2%	21.4%	Adj. EBITDA	6.0%
Appian	APPN	11.0%	9.4%	5.0%	Adj. EBITDA	6.0%
Anaplan	PLAN	30.3%	25.4%	25.6%	Adj. EBITDA	4.9%
Descartes	DSGX	65.5%	63.2%	62.1%	Adj. EBITDA	3.4%
Elastic	ESTC	47.7%	45.3%	44.8%	Adj. EBITDA	2.9%
Couchbase	BASE	-14.9%	-17.2%	-14.7%	Free Cash Flow	2.5%
MongoDB	MDB	51.0%	49.4%	47.9%	Adj. EBITDA	3.1%
Dynatrace	DT	59.9%	58.6%	59.3%	Adj. EBITDA	1.3%

TTM metrics used for all companies. Total revenue growth used for all companies as top-line growth measure. Green color indicates the column with the highest value for each respective company, yellow represents middle value, red indicates lowest value. Data sorted by "Differences from Best to Worst". ORCL FCF adj. by tax-affected litigation charge of $4.67bn; INTU revenue ex-Mailchimp and Credit Karma revenue; BIGC revenue ex-Feedonomics. CXM excluded due to reporting Q4 on 4/6.
Source: Barclays Research, Company Data

As seen in the figure above, calculated Rule of 40 can vary materially depending on the profitability margin used in the formula, and hence, investors must consider the various business factors that influence calculated Rule of 40. In the section below, we break down five key factors that investors should understand when using the metric to assess business health.

Factors Influencing Rule of 40

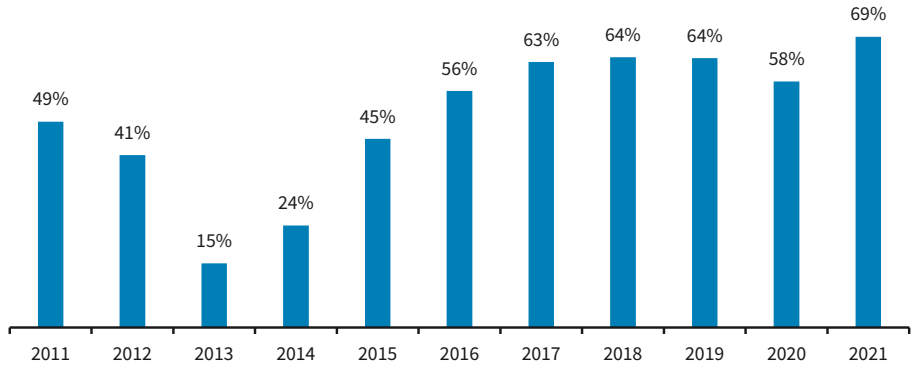
While using different proxies for top-line growth or profitability in the Rule is not inherently wrong, investors need to understand common factors that will affect the Rule of 40 interpretation. We discuss how deployment model, capital intensity, leverage profile, M&A and amortization methodology are all important factors that must be understood when using the Rule of 40.

Deployment Model

Subscription transitions in software have been a major theme for investors over the last decade with the vast majority of new software vendors coming to market with a cloud-first, modern SaaS solution. We have previously written about the impacts of a subscription transition on a company's financial statements (see A PM's Guide to Subscription Transitions, 4/5/18) - particularly highlighting the initial impact to both top-line growth and operating margin as the increased mix of ratable revenue recognition negatively affects top-line growth and margin in the short-term, but allows for better operating leverage exiting the transition.

Adobe (covered by Barclays Analyst Saket Kalia), which began its subscription transition in 2012, provides a helpful example of the financial impacts before, during and exiting a transition. As seen in the figure below, Adobe saw it's near 50% calculated Rule of 40 reduce to just 15% during its subscription transition. Interestingly, the company actually saw negative total revenue growth y/y in 2013, which does not always happen during a transition (typically only see a sharp deceleration instead of negative growth), but is possible if the migration to subscription solutions with ratable revenue recognition profile is done quickly. As seen in the figure below, Adobe exited the transition with better growth and operating leverage than pre-transition, and therefore, the sizable dip in Rule of 40 from 2012-2014 can be seen as not representative of the underlying health of the business. In these cases of subscription transitions, ARR growth may be an appropriate substitute for revenue growth, which helps normalize the revenue headwinds from increased ratable revenue recognition.

FIGURE 6. **Adobe's Subscription Transition Effect on Rule of 40 Evident in 2012-2015**



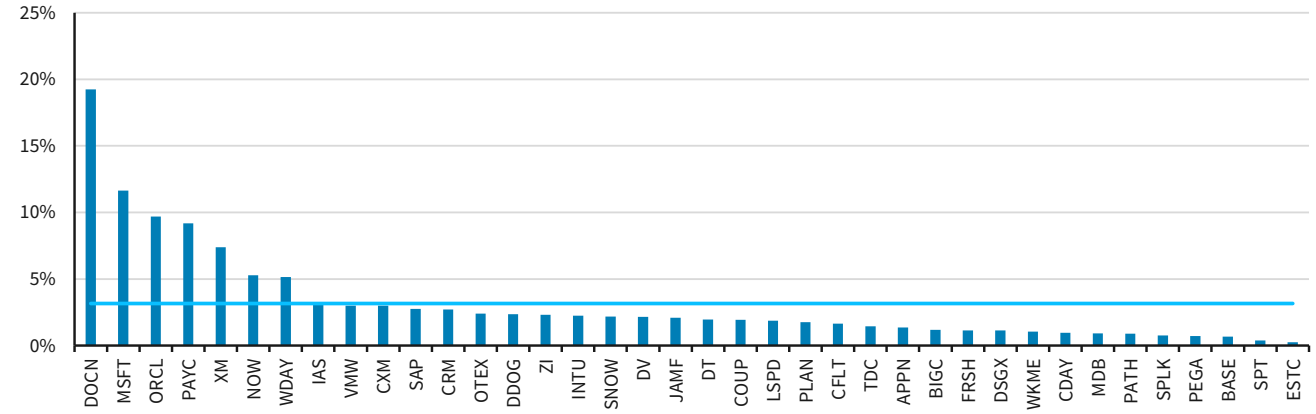
Rule of 40 calculated using total revenue growth + adj. operating margin
Source: Barclays Research,

Capital Intensity and M&A

Capital intensity, or the requirement of a higher proportion of fixed assets relative to other factors of production such as labor, can result in significantly different calculated Rule of 40 depending on what profitability metric is used (e.g. EBITDA vs. Operating Profit vs. FCF margin). While typically not as important to Software investors, compared to those who specialize in other sectors such as Energy, capital intensity plays an important role for firms that provide cloud infrastructure (e.g. Microsoft, DigitalOcean and Oracle) or chose to manage their own data

centers. When comparing capital intensive vendors to non-capital intensive vendors, it may be useful to use the Rule of 40 with adj. operating profit margin or FCF margin.

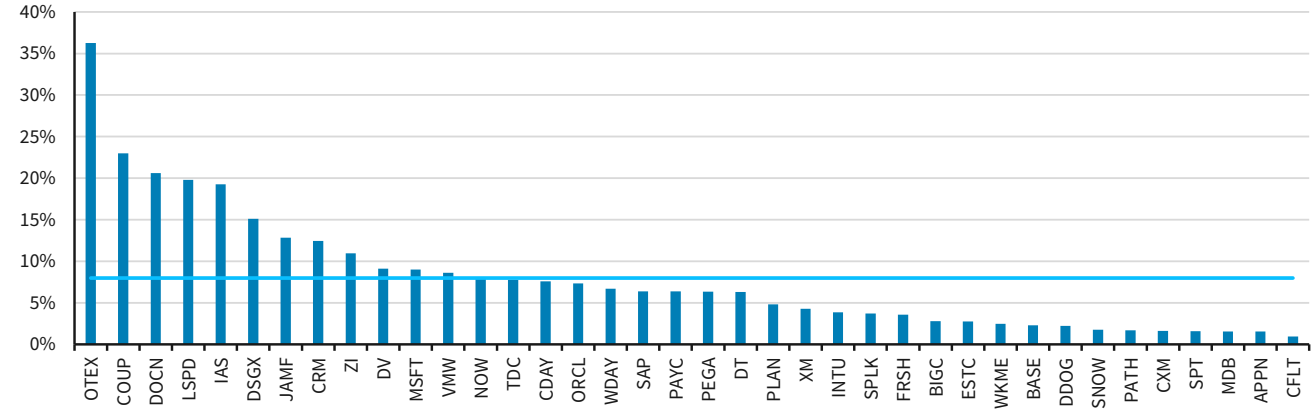
FIGURE 7. **Capex as a % of Revenue Across Coverage**



Light blue line indicates average of 3%. TTM metrics used for each company.
Source: Barclays Research, Company data

High levels of depreciation and amortization, often a byproduct of capital intensity, create a material delta between EBITDA and operating profit margin that can alter the investment conclusion in certain cases. However, high levels of D&A are also present for companies that have conducted M&A transactions. Examples here include Coupa, Lightspeed and Descartes.

FIGURE 8. **D&A as % of Revenue Across Coverage**



Light blue line indicates average of 8%. TTM metrics used for each company.
Source: Barclays Research, Company data

Leverage Profile

Financial leverage, specifically varying levels of debt servicing costs, is another factor that must be understood by investors given its impact on calculated Rule of 40. While not relevant for investors using EBITDA or operating profit margin when calculating the Rule of 40, higher interest expense lowers net income all else equal, and hence lowers free cash flow. In this sense, investors must understand the capital structure of target companies when using FCF in the Rule of 40 calculation. This is particularly important when comparing two companies with extreme variations in capital structure, where using unlevered FCF may serve as a better component vs. levered FCF in the formula.

Amortization of Commissions and Other

Amortization policy, specifically a longer duration in which companies recognize deferred contract acquisition costs, can create a P&L benefit that helps operating profit and EBITDA. As seen in the figure below, most companies recognize deferred costs over a period of five years, however, companies such as Paycom and OpenText amortize these costs over a slightly longer period, with Paycom amortizing these costs over ten years. All else equal, a longer amortization period creates a P&L benefit as long as companies are growing, particularly capitalizing these costs at a higher clip than amortizing these costs. This impact is particularly pronounced in the short-term when companies lengthen amortization periods for these costs, which creates a benefit from a y/y S&M cost perspective. In these cases, we think it is best to look at FCF margin in the Rule of 40 calculation for operational comparison and valuation purposes.

FIGURE 9. **Amortization Period for Deferred Contract Costs**

Company	Ticker	Deferred Costs Amortization Period
DigitalOcean	DOCN	N/A
Oracle	ORCL	4 Years
Paycom	PAYC	10 Years
Coupa	COUP	5 Years
Ceridian	CDAY	5 Years
OpenText	OTEX	6 Years
Microsoft	MSFT	N/A
Qualtrics	XM	5 Years
SAP	SAP	Expected contract life including possible renewals
DoubleVerify	DV	N/A
UiPath	PATH	5 Years
Lightspeed POS	LSPD	Consistent with transfer of good/service
Intuit	INTU	3-4 Years
Jamf	JAMF	5 Years
Splunk	SPLK	5 Years
BigCommerce	BIGC	Estimated period of our relationship with customers
Confluent	CFLT	5 Years
Sprout Social	SPT	3 Years
Snowflake	SNOW	5 Years
Datadog	DDOG	4 Years
Integral Ad Science	IAS	N/A
ServiceNow	NOW	5 Years
Teradata	TDC	4 Years
WalkMe	WKME	4 Years
ZoomInfo	ZI	3 Years
Workday	WDAY	5 Years
Freshworks	FRSH	3 Years
Salesforce	CRM	4 Years
VMware	VMW	Over Period of Expected Benefit
Pegasystems	PEGA	5 Years
Appian	APPN	5 Years
Anaplan	PLAN	Over Period of Expected Benefit
Descartes	DSGX	4-6 Years
Elastic	ESTC	5 Years
Couchbase	BASE	3 Years
MongoDB	MDB	5 Years
Sprinklr	CXM	3 Years
Dynatrace	DT	3 Years

N/A indicates amortization period not explicitly defined in filings.
Source: Company Data, Barclays Research

Using the Rule of 40 for Valuation

In our view, recent market volatility for growth assets has pushed investors to pursue the highest quality assets, which makes the Rule of 40 an increasingly useful metric given its ability to measure business health and the benefits of the rule discussed previously in this note. In this sense, we think the rule works well in comparing companies from an operational perspective, but our analysis (detailed later in this section) demonstrates that when looking at the Rule of 40 in the context of valuation, a modified Rule of 40, giving more weight to top-line growth versus profitability, has more explanatory power for valuation. In our analysis, we compare forward valuation multiples (EV/CY23E Sales) to the different definitions of the Rule of 40 as well as the modified approach. Interestingly, we found that using FCF in the Rule of 40 calculation generally correlates with valuation more closely than EBITDA or operating profit margin (see below). In our view, this makes sense as FCF takes into consideration factors such as capital intensity, leverage and amortization policy.

FIGURE 10. **Rule of 40 Using FCF Margin Correlates with Valuation Better than Operating or EBITDA Margin**

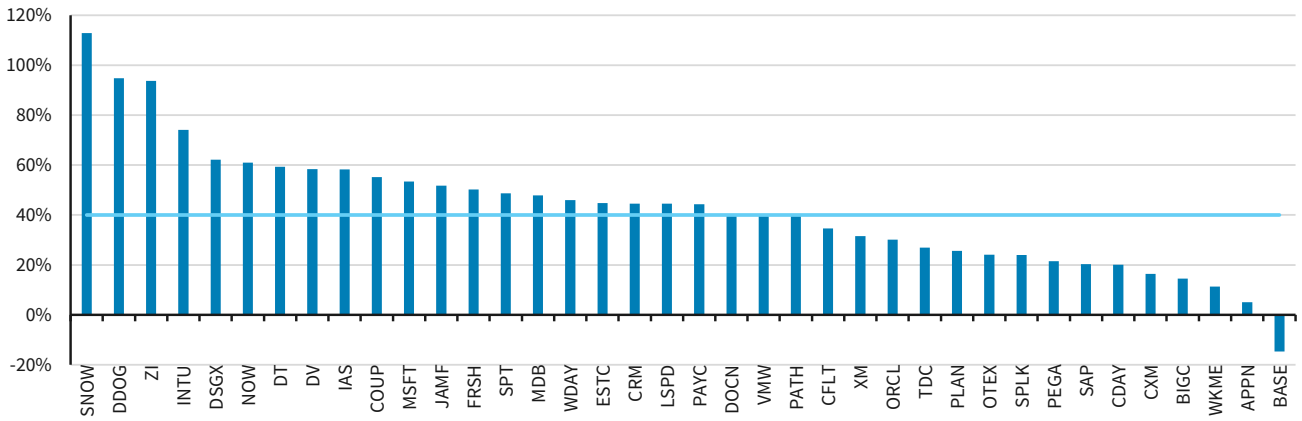
EV/CY23E Sales vs...	R-Squared
RO40: FCF Margin	52.38%
RO40: Operating Profit Margin	35.89%
RO40: EBITDA Margin	35.75%

TTM metrics and total revenue growth used for each company. RO40 = Rule of 40
Source: Barclays Research, Company data

Rule of 40 in the Context of Fundamentals

The simplicity and flexibility of the Rule of 40 allows investors to substitute the most appropriate top-line and profitability metrics into the calculation that is most representative of the underlying fundamentals of the business. The Rule of 40 also benefits from being relatively intuitive, making it convenient for competitor analysis. However, the most appropriate measure is not always straightforward, therefore, investors should consider the various factors described previously that impacts the rule's output. In our view, free cash flow overall is the superior profitability metric to use for profitability, which is supported by our analysis (in the next section). However, as noted, there are certain cases that lead to other metrics being more reflective of underlying business health. Looking across our coverage, we can see SNOW, DDOG, ZI and INTU stand out as particularly strong businesses (see figure below). While this shows the highest quality assets in our space, it does not take into account valuation.

FIGURE 11. **Rule of 40 Across our Coverage Using FCF Margin**

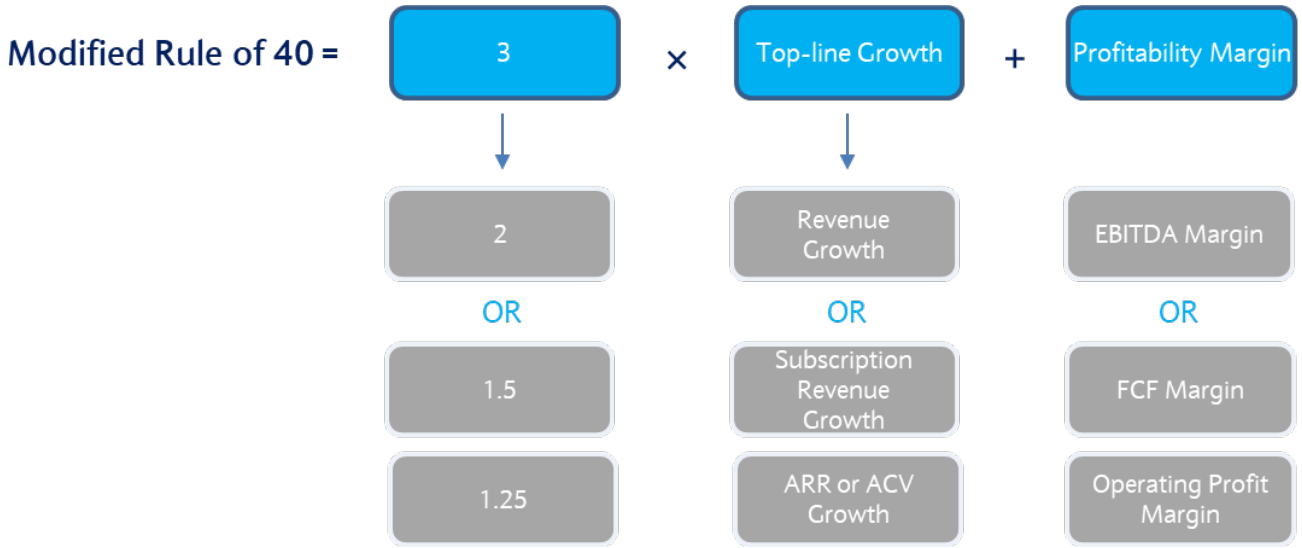


Light blue line = 40%. TTM metrics used for each company.
Source: Barclays Research, Company data

A Better Metric for Valuation? A Brief Intro to the Modified Rule of 40

As discussed, the Rule of 40 is helpful for analyzing business quality and comparing different companies, but software businesses, particularly those that are in the early stages of maturity, tend to value growth over profitability. The idea is that a healthy software business has a high gross margin that scales over time, and software vendors with a strong product/market fit should seek to take market share from competitors, innovate to establish a competitive moat, then, as growth scales and rising customer acquisition costs lower ROI, vendors should look to increase profitability and return capital to shareholders. This approach has led some venture capitalists to adopt a modified approach for using the rule for valuation, where greater weight is placed on the top-line component of the calculation by multiplying revenue growth (for example) by a factor of 1.25-3.0x, then adding the chosen profitability margin (see below).

FIGURE 12. **Modified Rule of 40 Values Top-line Growth More than Profitability**



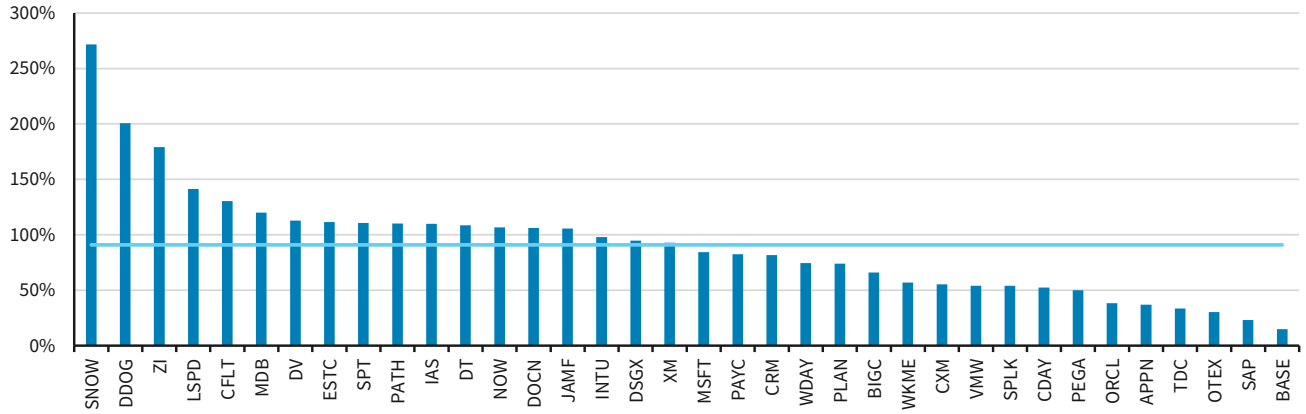
Source: Barclays Research

Ran Ding, partner at Norwest Venture Partners, discussed in a 2019 blog post that Norwest found that using a 3x multiple in the Rule of 40 leads to the highest correlation to valuation looking at public SaaS companies from 2016-2018 compared to the traditional Rule of 40 definition and forward sales growth (see Keeping it 100: A Better Way of Assessing SaaS

Performance than "Rule of 40"). Additionally, the analysis finds that the traditional definition was less useful as a predictor of valuation as indicated by decreasing R-squared values over time during the assessed period, supporting the idea that investors have increasingly valued top-line growth vs. margin and why most of our coverage is valued using EV/Sales.

While we think a 3x multiple may be appropriate framework for valuating software companies in early stages, our analysis shows that a modified approach using 1.5x revenue growth (+ FCF margin) leads to the highest correlation to current multiples for our coverage. Below, we present the modified Rule of 40 using 1.5x revenue growth + FCF margin for our coverage. While less intuitive than the traditional Rule of 40, we see the average modified Rule of 40 of 91% for our coverage as a helpful benchmark.

FIGURE 13. Modified Rule of 40 (1.5x Revenue Growth + FCF margin)

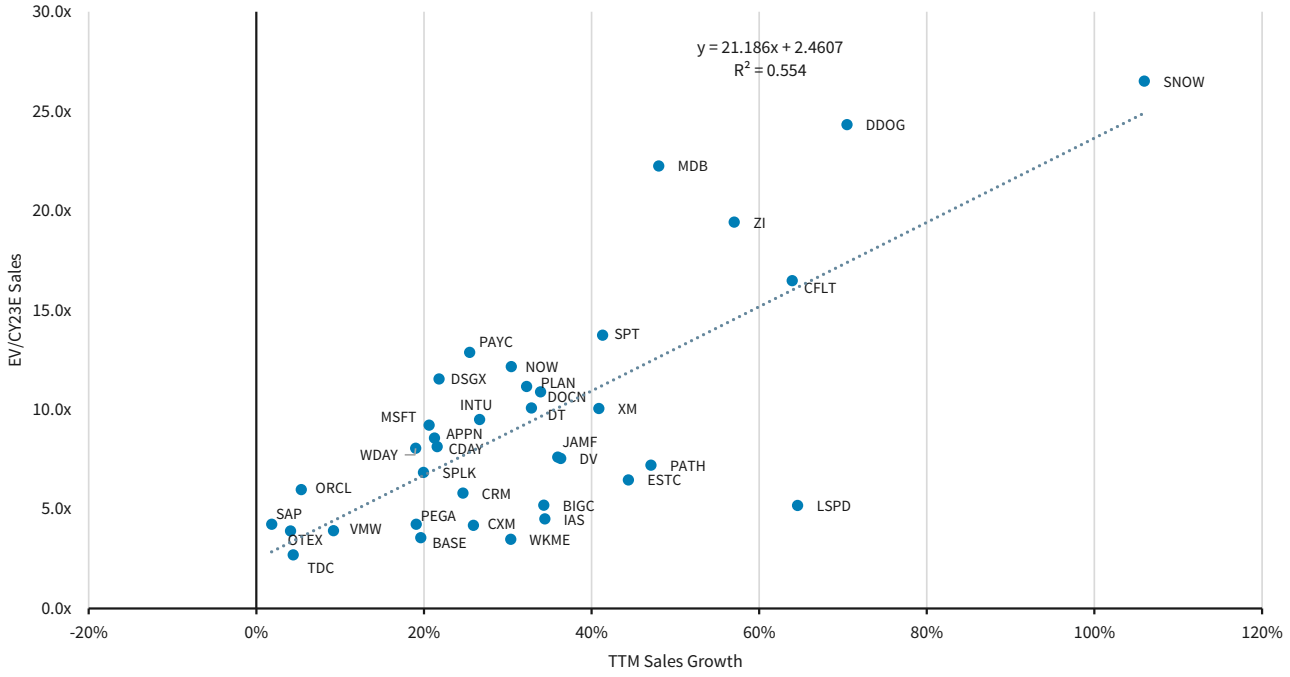


Light blue line indicates average of 91%. TTM metrics used for each company.
Source: Barclays Research, Company data

Rule of 40 in the Context of Valuation

To test the Rule of 40 as a driver of valuation, we compared current valuation levels using EV/CY23E Sales multiples to various definitions of the metric (e.g. using FCF vs. EBITDA margin in formula), as well as used the modified approach to give sales growth a higher weight in the formula, as described in the previous section. In particular, we tested 15+ variations of the metric, using a simple regression framework to measure R-squared values, and hence provide a sense of correlation. To provide a control measure, we looked at using only sales growth as a driver of valuation (see figure below). In our view, a higher r-squared value indicates a better fit, and therefore indicates the metric is a better driver of valuation. Below, we compare forward sales multiples for our coverage to sales growth and find an R-squared value of 55.40%.

FIGURE 14. EV/CY23E Sales vs. Sales Growth Creates Good Fit, but Can Do Better (R^2=55.40%)



Source: Barclays Research, Company data

As seen in the figure above, using TTM sales growth as a predictor of valuation yields a 55.40% R-squared value, which in our view is solid, but further analysis shows that we can approve the model fit by using the modified Rule of 40. In the figure below, we present the R-squared output for each the 15 tests conducted. Interestingly, the R-squared value for using the modified approach is higher than using only sales growth (the control) in every test we conducted, with FCF margin as the profitability metric in the tests with the top four outcomes.

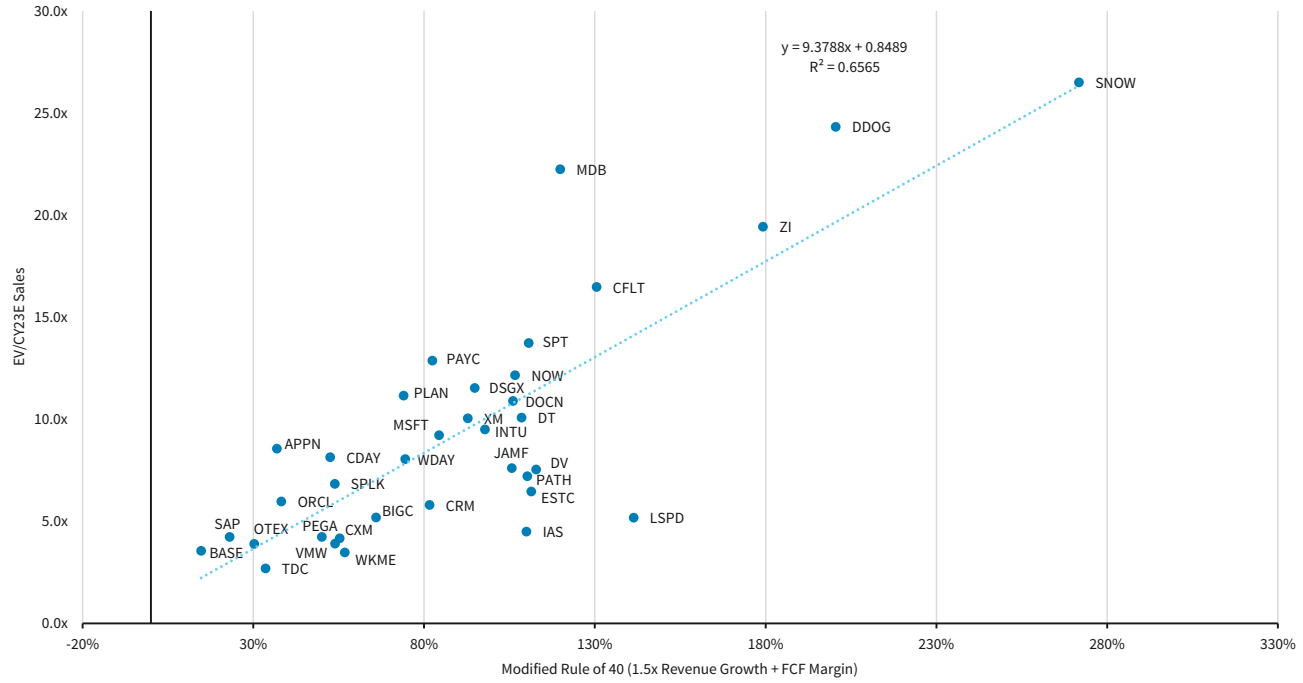
FIGURE 15. **Summary of EV/CY23E Sales vs. Variations of Rule of 40**

EV/CY23E Sales vs...	R^2
Modified RO40: (1.5x Revenue Growth + FCF Margin)	65.65%
Modified RO40: (1.25x Revenue Growth + FCF Margin)	65.64%
Modified RO40: (2x Revenue Growth + FCF Margin)	65.21%
Modified RO40: (2.5x Revenue Growth + FCF Margin)	64.56%
Modified RO40: (2.5x Revenue Growth + EBITDA Margin)	62.35%
Modified RO40: (2x Revenue Growth + EBITDA Margin)	62.24%
Modified RO40: (1.5x Revenue Growth + EBITDA Margin)	61.41%
Modified RO40: (2.5x Revenue Growth + Operating Margin)	61.28%
Modified RO40: (2x Revenue Growth + Operating Margin)	61.08%
Modified RO40: (1.25x Revenue Growth + EBITDA Margin)	60.45%
Modified RO40: (1.5x Revenue Growth + Operating Margin)	60.20%
Modified RO40: (1.25x Revenue Growth + Operating Margin)	59.24%
TTM Sales Growth Only (Control)	55.40%
RO40: FCF Margin	52.38%
RO40: Operating Profit Margin	35.89%
RO40: EBITDA Margin	35.75%

TTM metrics and total revenue growth used for each company. RO40 = Rule of 40.
Source: Barclays Research, Company data

Below, we present EV/CY23E sales multiples vs. the modified Rule of 40 (1.5 x revenue growth + FCF margin), which provided the highest R-squared value compared to the other tests. In our view, this supports that this modified approach should be considered when searching for underappreciated assets.

FIGURE 16. **Using Modified Rule of 40 (1.5 x Revenue Growth + FCF Margin) Leads to Best Fit Model**



TTM metrics used for each company. Total revenue growth used for each company.
Source: Barclays Research, Company data

Stocks We Like Using The Modified Rule of 40 Valuation Framework

From this framework and in the context of market dynamics, we think ESTC, CRM, and JAMF are underappreciated assets that deserve a second look. These names are examples of stocks that are "below the line" and hence seems to suggest a disconnect between the quality of the asset compared to what investors are willing to pay for it.

ESTC (OW rated): Elastic's disconnected valuation vs. the strong fundamentals, which was highlighted again in its recent print, remains a conundrum for us. The company ticked multiple boxes in its most recent print that may have been expected, including strong SaaS revenue growth (80% vs. 84% in Q2), reaccelerating billings growth (35% ccy vs. 29% in Q2), and strong traction from its >$100k ACV customers (33% y/y vs. 28% in Q2). What was debated before the print, however, was where operating margins might land, and here management provided positive clarification and set an expectation to invest in a disciplined manner to comfortably achieve over >$1bn of revenue in FY23. We feel the mostly positive data points reinforcing strong fundamentals and current valuation can make shares start to work.

CRM (OW rated): Salesforce remains our Top Pick and we believe that the recent Q4 print is what Salesforce needed to see shares recover from a less than ideal start to the calendar year. This quarter hit on the key items that investors needed to see in our view: Strong growth in cRPO and a beat to guidance, momentum building around Slack (large customer strength highlighted here), and a commitment to margin discipline and no material M&A in the near term. All of this, coupled with an attractive valuation and return to a beat and raise operating motion, should be enough to drive shares back higher, in our view.

JAMF (EW rated): We're growing more constructive on the company now that it has more opportunities for growth beyond just device expansion through further penetration into Apple's installed base. One of these growth opportunities is coming from stronger traction in the commercial markets (65%-70% of ARR), where ASPs per device are traditionally 5x higher the ASPs per device in education. Another is coming from a stronger upsell motion for security products. We like the growing breadth of the upsell motion with products such as Connect, Protect, Private Access, Threat Defense, and Data Policy, and see a long runway ahead with only ~8k customers out of over 60k customers using one or more of these security solutions. The one caveat is tougher comps for ARR and revenue growth and NRR are ahead in Q3, but with this well-messaged and the upselling motion gaining steam, the company stands to potentially navigate through this period.

Analyst(s) Certification(s):
I, Raimo Lenschow, CFA, hereby certify (1) that the views expressed in this research report accurately reflect my personal views about any or all of the subject securities or issuers referred to in this research report and (2) no part of my compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this research report.

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Materially Mentioned Stocks (Ticker, Date, Price)
Elastic N.V. (ESTC, 01-Apr-2022, USD 92.26), Overweight/Positive, A/CD/CE/D/J/L

Jamf Holding Corp. (JAMF, 01-Apr-2022, USD 34.42), Equal Weight/Positive, A/CD/CE/D/E/J/L

Salesforce.com Inc. (CRM, 01-Apr-2022, USD 212.25), Overweight/Positive, A/CD/CE/D/J/K/L/M

Unless otherwise indicated, prices are sourced from Bloomberg and reflect the closing price in the relevant trading market, which may not be the last available closing price at the time of publication.

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In addition to the stock rating, we provide industry views which rate the outlook for the industry coverage universe as Positive, Neutral or Negative (see definitions below). A rating system using terms such as buy, hold and sell is not the equivalent of our rating system. Investors should carefully read the entire research report including the definitions of all ratings and not infer its contents from ratings alone.

Stock Rating
Overweight - The stock is expected to outperform the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

Equal Weight - The stock is expected to perform in line with the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

Underweight - The stock is expected to underperform the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

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U.S. Software

8x8 Inc. (EGHT)
Adobe Inc. (ADBE)
Alarm.com Holdings, Inc. (ALRM)
Alkami Technology, Inc. (ALKT)
Anaplan, Inc. (PLAN)
Ansys, Inc. (ANSS)
Appian Corporation (APPN)
AudioCodes Ltd. (AUDC)
Autodesk Inc. (ADSK)
Avaya Holdings Corp (AVYA)
Bandwidth Inc. (BAND)
BigCommerce (BIGC)
Braze Inc. (BRZE)
Ceridian HCM Holding Inc. (CDAY)
Check Point Software Technologies Ltd. (CHKP)
Citrix Systems (CTXS)
Confluent, Inc (CFLT)
Couchbase (BASE)
Coupa Software Inc. (COUP)
CrowdStrike Holdings, Inc (CRWD)
CyberArk Software (CYBR)
Datadog, Inc. (DDOG)
Datto Holding Corp. (MSP)
Definitive Healthcare Corp (DH)
Descartes Systems Group (DSGX)
DigitalOcean (DOCN)
DoubleVerify Holdings, Inc. (DV)
Duck Creek Technologies, Inc. (DCT)
Dynatrace, Inc. (DT)
Elastic N.V. (ESTC)
Everbridge, Inc. (EVBG)
EverCommerce Inc. (EVCM)
Five9, Inc. (FIVN)
Fortinet, Inc. (FTNT)
Freshworks Inc. (FRSH)
HubSpot, Inc. (HUBS)
Integral Ad Science Holding Corp. (IAS)
Intuit Inc. (INTU)
Jamf Holding Corp. (JAMF)
Lightspeed Commerce Inc. (LSPD)
LivePerson, Inc. (LPSN)
Mandiant, Inc. (MNDT)
MeridianLink, Inc. (MLNK)
Microsoft Corp. (MSFT)
Mimecast Ltd. (MIME)
MongoDB, Inc. (MDB)
nCino, Inc. (NCNO)
NortonLifeLock (NLOK)
Open Text Corp. (OTEX)
Oracle Corp. (ORCL)
Palo Alto Networks (PANW)
Paycom (PAYC)
Pegasystems, Inc. (PEGA)
Ping Identity Holding Corp. (PING)
PowerSchool Holdings, Inc (PWSC)
Procore Technologies, Inc. (PCOR)
PTC Inc. (PTC)
Qualtrics (XM)
Rapid7 (RPD)
RingCentral, Inc. (RNG)
Salesforce.com Inc. (CRM)
SAP SE (SAP)
SecureWorks (SCWX)
SentinelOne, Inc. (S)
ServiceNow, Inc. (NOW)
Similarweb Ltd. (SMWB)
Skillsoft Corp. (SKIL)
Snowflake Computing (SNOW)
Splunk Inc. (SPLK)
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UiPath, Inc. (PATH)
Varonis Systems, Inc. (VRNS)
Veeva Systems Inc. (VEEV)
VMware Inc. (VMW)
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WalkMe Ltd. (WKME)
Workday Inc. (WDAY)
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Elastic N.V. (ESTC / ESTC)

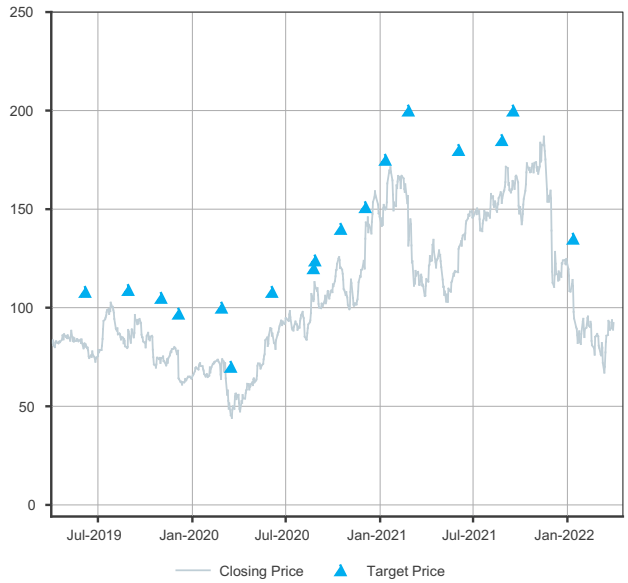
Stock Rating: **OVERWEIGHT**

Industry View: **POSITIVE**

USD 92.26 (01-Apr-2022)

Rating and Price Target Chart - USD (as of 01-Apr-2022)

Currency=USD



Source: IDC, Barclays Research

Link to Barclays Live for interactive charting

Publication Date	Closing Price*	Rating	Adjusted Price Target
12-Jan-2022	112.01		135.00
17-Sep-2021	159.99		200.00
26-Aug-2021	153.15		185.00
03-Jun-2021	129.84		180.00
25-Feb-2021	131.56		200.00
11-Jan-2021	149.88		175.00
03-Dec-2020	135.00		151.00
16-Oct-2020	120.48		140.00
27-Aug-2020	111.16		124.00
24-Aug-2020	103.35		120.00
04-Jun-2020	85.85		108.00
16-Mar-2020	45.37		70.00
27-Feb-2020	71.93		100.00
05-Dec-2019	64.22		97.00
01-Nov-2019	74.38		105.00
29-Aug-2019	88.76		109.00
06-Jun-2019	80.25		108.00

On 04-Apr-2019, prior to any intra-day change that may have been published, the rating for this security was Overweight, and the adjusted price target was 107.00.

Source: Bloomberg, Barclays Research

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Valuation Methodology: Our price target of $135 is based on ~10x CY23E revenue of ~$1.30bn.

Risks which May Impede the Achievement of the Barclays Research Valuation and Price Target: Elastic is expanding into the observability and security markets where it faces more competition, and if traction of these solutions slows, or if uptake of Elastic Cloud declines, it could adversely impact revenue growth.

Jamf Holding Corp. (JAMF / JAMF)

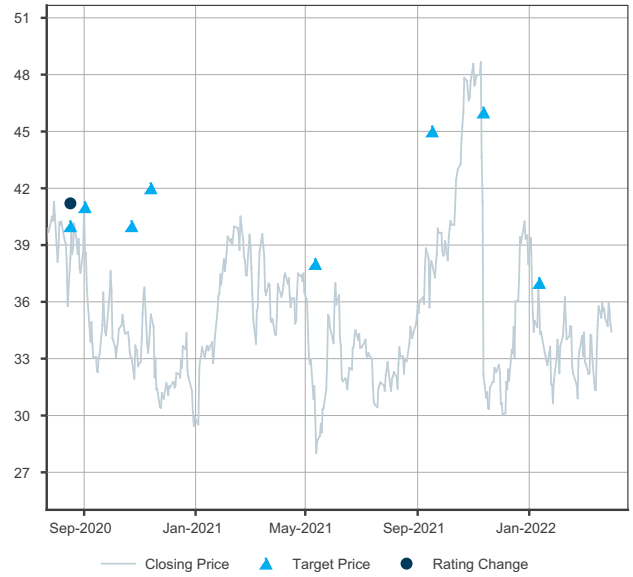
Stock Rating: **EQUAL WEIGHT**

Industry View: **POSITIVE**

USD 34.42 (01-Apr-2022)

Rating and Price Target Chart - USD (as of 01-Apr-2022)

Currency=USD



Source: IDC, Barclays Research

Link to Barclays Live for interactive charting

Publication Date	Closing Price*	Rating	Adjusted Price Target
12-Jan-2022	34.67		37.00
12-Nov-2021	32.14		46.00
17-Sep-2021	37.90		45.00
12-May-2021	29.86		38.00
13-Nov-2020	35.33		42.00
23-Oct-2020	33.02		40.00
02-Sep-2020	38.50		41.00
17-Aug-2020	38.33	Equal Weight	40.00

Source: Bloomberg, Barclays Research

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Valuation Methodology: Our PT of $37 is based on ~28x EV/uFCF and CY23E uFCF of $173mn.

Risks which May Impede the Achievement of the Barclays Research Valuation and Price Target: The company may see better than expected adoption driven by secular tailwinds and benign competitive landscape, which may lead to accelerating growth prospects and meaningful upside to shares. On the downside, tougher competitive landscape, disruption to Apple partnership or weaker than expected momentum for Apple in the enterprise may pressure growth prospects.

Salesforce.com Inc. (CRM / CRM)

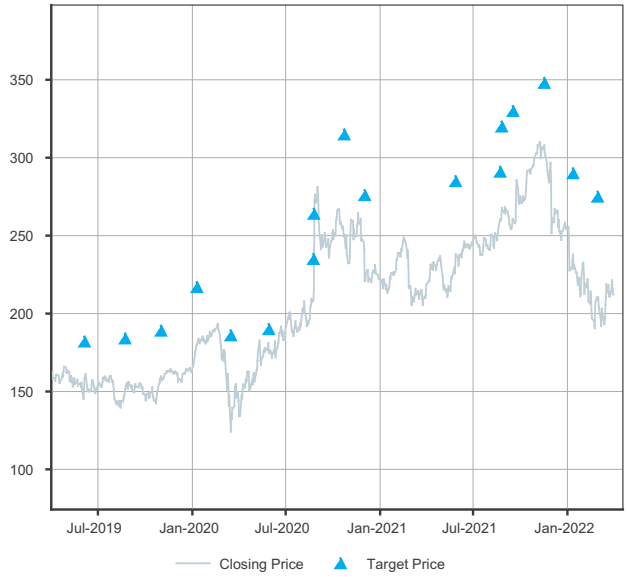
Stock Rating: **OVERWEIGHT**

Industry View: **POSITIVE**

USD 212.25 (01-Apr-2022)

Rating and Price Target Chart - USD (as of 01-Apr-2022)

Currency=USD



Source: IDC, Barclays Research

Link to Barclays Live for interactive charting

Publication Date	Closing Price*	Rating	Adjusted Price Target
01-Mar-2022	208.89		275.00
12-Jan-2022	229.63		290.00
17-Nov-2021	305.49		348.00
17-Sep-2021	256.16		330.00
26-Aug-2021	267.79		320.00
23-Aug-2021	260.52		291.00
28-May-2021	238.10		285.00
02-Dec-2020	220.78		276.00
23-Oct-2020	250.52		315.00
25-Aug-2020	216.05		264.00
24-Aug-2020	208.46		235.00
29-May-2020	174.79		190.00
16-Mar-2020	124.30		186.00
10-Jan-2020	180.20		217.00
01-Nov-2019	159.74		189.00
23-Aug-2019	151.57		184.00

05-Jun-2019	158.44	182.00

On 04-Apr-2019, prior to any intra-day change that may have been published, the rating for this security was Overweight, and the adjusted price target was 180.00.

Source: Bloomberg, Barclays Research

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Valuation Methodology: Our price target of $275 is based on 32x CY23E EV/FCF and CY23E FCF of $8.779bn

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